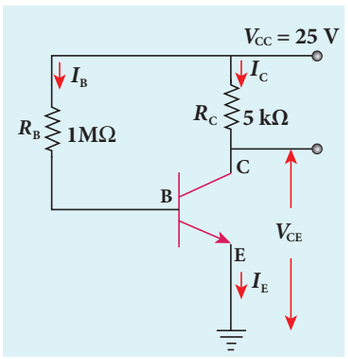
- **When the input is high:**

When the input voltage is increased to a certain high value (say +5 V), the base current (I_B) increases and in turn increases the collector current to its maximum. The transistor will move into the saturation region. The increase in collector current (I_C) increases the voltage drop across R_C, thereby lowering the output voltage, close to zero (since $V_0 = V_{CC} - I_C R_C$). It means that maximum current flows through the transistor and it is said to be switched on. The transistor acts as a closed switch.

It is manifested that a high input to the transistor gives a low output and a low input gives a high output. In addition, we can say that the output voltage is opposite to the applied input voltage. Therefore, a transistor can be used as an inverter (NOT gate) in computer logic circuitry.

EXAMPLE 10.7

The current gain of a common emitter transistor circuit shown in figure is 120. Draw the DC load line and mark the Q point on it. (V_{BE} to be ignored).



Solution

$\beta = 120$

Base current, $I_B = \dfrac{25V}{1M\Omega} = \dfrac{25}{1\times10^6}$

$= 25\ \mu A$

We know that

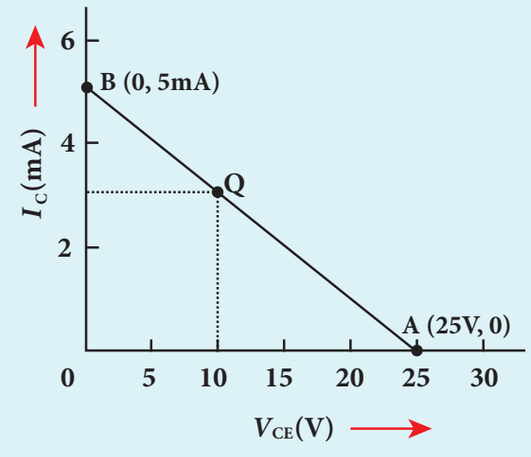
$\beta = \dfrac{I_C}{I_B}$ (or)

$I_C = \beta I_B = 120 \times 25\ \mu A$

$= 3000\ \mu A = 3\ mA$

$V_{CE} = V_{CC} - I_C R_C$

$= 25 - (3mA \times 5k) = 10\,V$



10.4.7 Transistor as an amplifier

A transistor operating in the active region has the capability to amplify weak signals. **Amplification is the process of increasing the signal strength (increase in the amplitude).** If a large amplification is required, the transistors are cascaded with coupling elements like resistors, capacitors, and transformers and they are called multistage amplifiers.

Here, the amplification of an electrical signal is explained with a single stage transistor amplifier which is shown in Figure 10.36(a). Single stage indicates that the circuit consists of one transistor with the allied components. An NPN transistor is connected in the common emitter configuration.

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